

Name: _____ Date: _____

Period: _____

Conic Sections Practice

Directions: Show all work. Circle your final answers.

1. For each equation, identify the conic type, center or vertex, and all key measurements.

(a) $y + 1 = \frac{1}{2}(x - 3)^2$

Type: _____ Vertex: _____ $a =$ _____ Opens: _____

(b) $\frac{(x + 2)^2}{36} + \frac{(y - 4)^2}{16} = 1$

Type: _____ Center: _____ $a =$ _____ $b =$ _____

(c) $\frac{(x - 5)^2}{4} - \frac{(y + 3)^2}{25} = 1$

Type: _____ Center: _____ $a =$ _____ $b =$ _____

2. Write the standard form equation for each description.

(a) A parabola with vertex $(0, 4)$ opening downward, passing through $(2, 0)$.

(b) A parabola with vertex $(3, -1)$ opening to the left, passing through $(1, 3)$.

(c) An ellipse centered at $(-1, 2)$ with horizontal radius 6 and vertical radius 4.

3. A hyperbola is centered at $(4, -3)$ with $a = 5$ and $b = 2$, opening up and down. Write the standard form equation and both asymptote equations.

4. The equation $9x^2 - 4y^2 = 36$ can be rewritten in standard hyperbola form by dividing both sides by 36.

(a) Rewrite in standard form. _____

(b) Identify the center, a , b , orientation, and write the asymptote equations. _____

AP-Style Practice

5. Which of the following is the standard form of a hyperbola with center $(2, -1)$, opening left and right, with $a = 3$ and $b = 4$?

(A) $\frac{(x-2)^2}{9} + \frac{(y+1)^2}{16} = 1$

(B) $\frac{(x-2)^2}{9} - \frac{(y+1)^2}{16} = 1$

(C) $\frac{(y+1)^2}{16} - \frac{(x-2)^2}{9} = 1$

(D) $\frac{(x+2)^2}{9} - \frac{(y-1)^2}{16} = 1$

Extension optional

Extension (optional): The equation $x^2 + 4y^2 - 6x + 8y - 3 = 0$ is a general-form ellipse. Complete the square in both x and y to rewrite it in standard form. Identify the center, horizontal radius, and vertical radius.